

Non-Destructive Estimation of Sugar Content in Fruits Using Visible-Light Imaging and Deep Learning

A Computer Vision Project Report
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Abstract

This paper presents **FruitSugar AI**, a non-destructive computer vision system for estimating sugar content in fruits using visible-light imaging and deep learning. Traditional sugar measurement techniques such as Brix refractometry are inherently destructive — requiring the fruit to be cut and juiced — making them impractical for large-scale sorting, consumer-level assessment, and quality-sensitive supply chains. Our proposed system addresses this challenge by leveraging **transfer learning** with a **MobileNetV2** convolutional neural network pre-trained on ImageNet, combined with a multi-modal feature extraction pipeline encompassing **HSV/LAB color histograms**, **Gray-Level Co-occurrence Matrix (GLCM) texture features**, and **Local Binary Patterns (LBP)**. The system classifies fruit images into three sugar content categories: *Low Sugar*, *Medium Sugar*, and *High Sugar*.

Trained on over **14,600 images** spanning four fruit types (apple, banana, orange, and pomegranate) from the FruitNet Indian Fruits Dataset, the model achieves an overall test accuracy of **99.68%**, with weighted precision, recall, and F1-score all at **99.68%**. The Medium Sugar class achieved **perfect classification (100.0%)** on the test set. A production-ready **Streamlit web application** provides real-time inference with confidence scores, estimated Brix ranges, and explainable feature visualizations, making the system accessible to non-technical users.

Keywords: Computer Vision, Deep Learning, Transfer Learning, MobileNetV2, Sugar Content Estimation, Non-Destructive Testing, Fruit Quality Assessment, CNN, GLCM, LBP, Color Histograms

1. Introduction

1.1 Background

The sugar content of fruits — typically measured in degrees Brix (°Bx) — is one of the most critical quality indicators in the agricultural and food processing industries. Sugar content directly influences taste, consumer satisfaction, market pricing, and shelf life. Globally, the fresh fruit market is valued at over \$500 billion, and sugar content grading is fundamental to the supply chain from farm to consumer.

The established gold-standard method for measuring sugar content is **optical refractometry**, which measures the refractive index of fruit juice. While accurate, this method is:

- **Destructive:** The fruit must be physically cut open and juiced
- **Time-consuming:** Each measurement takes manual effort
- **Wasteful:** Tested fruits cannot be sold
- **Not scalable:** Impractical for inline sorting at industrial speeds
- **Inaccessible:** Requires specialized equipment not available to consumers

These limitations create a significant need for **non-destructive estimation** methods that can rapidly assess sugar content from external visual characteristics alone.

1.2 Motivation

Recent advances in deep learning and computer vision have demonstrated remarkable success in visual recognition tasks. Convolutional Neural Networks (CNNs), particularly when combined with transfer learning, can extract subtle visual features from images that correlate with internal fruit properties. Research in food science has established that external visual characteristics — including skin color, surface texture, and ripeness indicators — are strongly correlated with internal sugar content:

- **Color changes** during ripening reflect chlorophyll degradation and carotenoid/anthocyanin accumulation, which co-occur with sugar synthesis
- **Surface texture** changes as cell walls soften during ripening, coinciding with starch-to-sugar conversion
- **Glossiness and sheen** variations correlate with wax layer changes during maturation

These biological relationships provide the scientific basis for our computer vision approach: if human experts can estimate ripeness (and by extension, approximate sugar content) by visual inspection, a deep neural network can be trained to perform this task with greater consistency and precision.

1.3 Contributions

The key contributions of this work are:

1. **End-to-end pipeline** for non-destructive sugar content estimation from standard visible-light photographs
2. **Multi-modal feature analysis** combining deep learning with handcrafted features (color histograms, GLCM, LBP) for explainability
3. **High-accuracy classification** achieving 99.68% accuracy across three sugar content levels
4. **Production-ready web application** with real-time inference, Brix range estimation, and visual feature explanations
5. **Comprehensive evaluation** with confusion matrices, per-class metrics, and training convergence analysis

2. Literature Review

2.1 Near-Infrared (NIR) Spectroscopy Methods

NIR spectroscopy is the most extensively studied non-destructive technique for sugar content measurement. Nicolaï et al. (2007) provided a comprehensive review of NIR applications in fruit quality assessment, demonstrating R^2 values of 0.85–0.95 for Brix prediction in apples. However, NIR systems are expensive (\$10,000–\$50,000), require calibration, and are not portable for consumer use.

2.2 Hyperspectral Imaging

Hyperspectral imaging captures hundreds of spectral bands and has been applied to sugar content prediction by Lu et al. (2017) with promising results. While accurate, hyperspectral cameras remain prohibitively expensive for widespread deployment.

2.3 Visible-Light Computer Vision Approaches

Several researchers have explored standard RGB camera-based approaches:

- **Zheng et al. (2020)** used color features and SVMs to classify apple ripeness stages, achieving 87% accuracy
- **Naik & Patel (2017)** employed texture-based features (GLCM and LBP) for mango quality grading with 89% accuracy
- **Wang et al. (2019)** applied CNN-based approaches to banana ripeness classification, achieving 92% accuracy with VGG16

2.4 Deep Learning for Fruit Quality Assessment

Transfer learning with pre-trained CNNs has emerged as the dominant paradigm:

- **Steinbrener et al. (2019)** fine-tuned InceptionV3 for apple defect detection, achieving 94% accuracy
- **Jahanbakhshi et al. (2020)** used AlexNet for pomegranate quality classification with 96% accuracy
- **Rauf et al. (2019)** employed MobileNet for real-time fruit recognition with 95% accuracy, demonstrating the architecture's suitability for edge deployment

2.5 MobileNetV2 Architecture

MobileNetV2, introduced by Sandler et al. (2018), builds upon MobileNetV1 by incorporating **inverted residual blocks** with **linear bottlenecks**. Key innovations include:

- **Depthwise separable convolutions** reducing computational cost by 8–9× compared to standard convolutions
- **Inverted residual connections** that shortcut between thin bottleneck layers
- **Linear bottlenecks** that prevent information loss from non-linear activations in low-dimensional spaces

These design choices make MobileNetV2 ideal for deployment scenarios requiring both high accuracy and low computational overhead — precisely our use case for real-time fruit assessment.

2.6 Research Gap

While existing literature demonstrates the feasibility of vision-based fruit quality assessment, most studies:

- Focus on a single fruit type
- Use expensive imaging equipment (NIR, hyperspectral)
- Lack a deployable application
- Do not provide explainability through handcrafted features

Our work addresses these gaps by building a **multi-fruit, standard-camera, deployable, and explainable** system.

3. Problem Statement & Objectives

3.1 Problem Statement

"To develop a non-destructive, cost-effective, and scalable system that can accurately estimate the sugar content level (Low, Medium, or High) of common fruits using only visible-light photographs, leveraging deep learning and computer vision techniques."

3.2 Objectives

#	Objective	Status
1	Collect and preprocess a large-scale fruit image dataset with quality labels	■ Achieved
2	Design a quality-to-sugar content mapping grounded in food science	■ Achieved
3	Implement a CNN-based classification model using transfer learning	■ Achieved
4	Extract handcrafted visual features for model explainability	■ Achieved
5	Achieve classification accuracy exceeding 90%	■ 99.68%
6	Develop an interactive web application for real-time inference	■ Achieved
7	Provide estimated Brix ranges based on food science literature	■ Achieved

4. System Architecture

4.1 High-Level Architecture Diagram

[Architecture diagram - see the .md file for interactive Mermaid diagrams]

4.2 Software Architecture

[Architecture diagram - see the .md file for interactive Mermaid diagrams]

4.3 Module Dependency Map

Module	Depends On	Purpose
config.py	—	Central configuration (paths, hyperparameters, constants)
utils.py	config	Helper functions (seed, device, formatting)
setup_dataset.py	config, utils	Dataset download, scanning, organization
dataset_utils.py	config	Custom Dataset class, DataLoaders, transforms
model.py	config, utils	MobileNetV2 model definition with custom head
feature_extraction.py	config, utils	Color, texture, and LBP feature extraction
train.py	config, utils, dataset_utils, model	Training loop with early stopping
evaluate.py	config, utils, dataset_utils, model	Test evaluation, metrics, visualizations
inference.py	config, model, feature_extraction, dataset_utils	Single/batch prediction with features

app.py	config, model, inference, feature_extraction	Streamlit web application
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5. Methodology

5.1 Dataset Description

5.1.1 Source

The project uses the **FruitNet: Indian Fruits Dataset with Quality**, publicly available on Kaggle (Shashwat, 2022). The dataset is licensed under **CC BY-SA 4.0**.

- **Kaggle URL:**
<https://www.kaggle.com/datasets/shashwatwork/fruitnet-indian-fruits-dataset-with-quality>

5.1.2 Dataset Characteristics

Property	Value
Total Images	14,640
Selected Fruit Types	4 (Apple, Banana, Orange, Pomegranate)
Original Quality Labels	Good, Bad, Mixed
Image Format	JPEG, PNG
Color Channels	RGB (3 channels)

5.1.3 Quality-to-Sugar Content Mapping

A key design decision in this project is the mapping from the dataset's original quality labels to sugar content estimation classes. This mapping is grounded in food science research:

Dataset Quality Label	Sugar Content Class	Food Science Rationale
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Good Quality	High Sugar ■	Optimally ripe fruits exhibit peak sweetness due to complete starch-to-sugar conversion. Visual indicators include vibrant coloring, uniform skin texture, and appropriate firmness.
Mixed Quality	Medium Sugar ■	Partially ripe fruits show intermediate sugar levels. Visual cues include partial color development and transitional texture characteristics.
Bad Quality	Low Sugar ■	Unripe fruits contain predominantly starch (low sweetness), while overripe fruits may show fermentation-related sugar breakdown. Both exhibit degraded visual quality.

■■ **Important:** Alternative naming conventions in the dataset (e.g., "fresh" → High Sugar, "rotten" → Low Sugar) are also handled through the mapping dictionary in config.py.

5.1.4 Data Split Strategy

The dataset is divided into three stratified splits, maintaining class proportions across all splits:

Split	Ratio	Total Images	High Sugar	Medium Sugar	Low Sugar
Training	70%	10,246	6,592	453	3,201
Validation	15%	2,195	1,412	97	686
Test	15%	2,199	1,414	98	687

Total	100%	14,640	9,418	648	4,574
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5.1.5 Class Distribution Analysis

The dataset exhibits significant **class imbalance**:

- High Sugar: **64.3%** of total samples (majority class)
- Low Sugar: **31.2%** of total samples
- Medium Sugar: **4.4%** of total samples (minority class)

This imbalance is addressed through **Weighted Random Sampling** during training (Section 5.7).

5.1.6 Estimated Brix Ranges (Based on Food Science Literature)

Fruit	High Sugar (°Bx)	Medium Sugar (°Bx)	Low Sugar (°Bx)
■ Apple	13–16	10–13	6–10
■ Banana	20–25	14–20	5–14
■ Orange	11–15	8–11	5–8
■ Pomegranate	15–19	12–15	8–12

5.2 Data Preprocessing & Augmentation

5.2.1 Preprocessing Pipeline

- All images undergo the following preprocessing steps:
1. **Resizing:** Images are resized to **224 × 224 pixels** (the standard input size for MobileNetV2)
 2. **Tensor Conversion:** PIL Images are converted to PyTorch tensors (pixel values normalized from 0–255 to 0.0–1.0)
 3. **ImageNet Normalization:** Pixel values are normalized using ImageNet statistics:
 4. Mean: **[0.485, 0.456, 0.406]** (per RGB channel)
 5. Std: **[0.229, 0.224, 0.225]** (per RGB channel)

5.2.2 Data Augmentation (Training Only)

To improve model generalization and prevent overfitting, the following augmentations are applied **only during training**:

Augmentation	Parameters	Purpose
Resize + Random Crop	Resize to 256×256, then random crop to 224×224	Spatial variety, simulates different framings
Random Horizontal Flip	p = 0.5	Orientation invariance
Random Vertical Flip	p = 0.2	Additional orientation variety
Random Rotation	±15°	Rotation invariance
Color Jitter — Brightness	±0.2	Lighting condition robustness
Color Jitter — Contrast	±0.2	Camera/display variation robustness
Color Jitter — Saturation	±0.2	Color intensity robustness
Color Jitter — Hue	±0.1	Slight color shift robustness

■ ■ **Note:** ****Color Jitter is Critical:**** Since our model relies heavily on color features to distinguish sugar content levels, the color jitter augmentation is carefully calibrated. The hue jitter (±0.1) is intentionally kept small to preserve the fundamental color information that correlates with sugar content, while still introducing enough variation for robustness.

5.2.3 Validation/Test Preprocessing

For validation and test sets, only deterministic transforms are applied:

1. Resize to 224 × 224 (no random crop)
2. Tensor conversion
3. ImageNet normalization

This ensures **consistent and reproducible** evaluation.

5.3 Feature Extraction Pipeline

In addition to deep learning-based classification, the system extracts three categories of handcrafted visual features for **explainability and analysis**. These features help users understand *why* the model makes specific predictions.

5.3.1 Color Histograms (HSV & LAB Color Spaces)

Why not RGB? While RGB is the standard representation, it mixes luminance and chrominance information. HSV and LAB color spaces provide better separation:

HSV (Hue, Saturation, Value):

- **Hue:** Represents the dominant wavelength (color type) — changes dramatically during ripening
- **Saturation:** Represents color purity/intensity — ripe fruits often have higher saturation
- **Value:** Represents brightness — useful for detecting browning or overripeness

LAB (Lightness, A-channel, B-channel):

- **L (Lightness):** Perceptual brightness, independent of color
- **A (Green↔Red axis):** Captures the green-to-red transition during ripening
- **B (Blue↔Yellow axis):** Captures yellowing associated with ripening

Implementation: 64-bin normalized histograms are computed for each channel using OpenCV's `cvtColor()` function.

5.3.2 GLCM Texture Features

The **Gray-Level Co-occurrence Matrix (GLCM)** captures spatial relationships between pixel intensities, providing a statistical description of surface texture.

Computation Parameters:

- Gray level quantization: 256 → 32 levels
- Distance: 1 pixel (immediate neighbors)
- Angles: 0°, 45°, 90°, 135° (rotation-invariant through averaging)
- Symmetric and normalized

Extracted Properties:

Feature	Description	Sugar Content Correlation
Contrast	Intensity difference between neighbors	Lower in smooth, ripe (high-sugar) fruits
Energy	Texture uniformity (Angular Second Moment)	Higher in uniformly textured ripe fruits

Homogeneity	Closeness of GLCM values to diagonal	Higher in smooth-skinned ripe fruits
Correlation	Linear dependency of gray levels	Indicates structural patterns

5.3.3 Local Binary Patterns (LBP)

LBP captures **micro-texture** patterns by comparing each pixel with its circular neighborhood:

Parameters:

- Radius: 2 pixels
- Number of neighbor points: 16
- Method: Uniform (rotation-invariant)
- Output: Histogram of 18 bins (16 uniform + 2 non-uniform patterns)

Statistics extracted: LBP histogram, mean, and standard deviation.

Why LBP matters for sugar estimation: Different ripeness levels produce distinct micro-texture signatures — unripe fruits often have tighter, more uniform cell structures visible as smoother textures, while ripe fruits may show more varied surface patterns.

5.4 Model Architecture — MobileNetV2

5.4.1 Architecture Choice Justification

MobileNetV2 was selected for the following reasons:

Criterion	MobileNetV2 Advantage
Accuracy	Achieves 72.0% top-1 on ImageNet (comparable to larger models)
Size	~3.4M parameters (vs. 138M for VGG16, 25.6M for ResNet50)
Speed	Optimized for real-time inference on mobile/edge devices
Transfer Learning	Strong feature representations from ImageNet pre-training

Deployment

Small model footprint (~12 MB) suitable for web deployment

5.4.2 MobileNetV2 Core Innovations

[Architecture diagram - see the .md file for interactive Mermaid diagrams]

Key Design Principles:

- 1. **Depthwise Separable Convolutions:** Factorize standard convolutions into depthwise (spatial filtering) and pointwise (channel mixing) operations, reducing computation by ~8-9×
- 2. **Inverted Residuals:** Unlike traditional residual blocks (wide→narrow→wide), MobileNetV2 uses narrow→wide→narrow structure
- 3. **Linear Bottlenecks:** The final 1×1 convolution uses a linear activation (no ReLU) to prevent information loss in low-dimensional representations

5.4.3 Custom Classifier Head

The original MobileNetV2 classifier (1280 → 1000 for ImageNet) is replaced with a custom multi-layer classifier:



Design rationale:

- **Two hidden layers** (512, 128) provide sufficient capacity for the 3-class task
- **BatchNorm** layers accelerate convergence and add regularization
- **Graduated Dropout** (0.3 → 0.15) applies stronger regularization earlier in the classifier
- **ReLU activations** are used throughout for computational efficiency

5.4.4 Model Parameters

Parameter Category	Count	Percentage
Total Parameters	~3,538,947	100%
Frozen (backbone)	~2,230,272	~63%
Trainable (unfrozen backbone + classifier)	~1,308,675	~37%

5.5 Transfer Learning Strategy

Transfer learning is the key technique enabling high accuracy with a relatively small, domain-specific dataset.

5.5.1 Strategy: Partial Fine-Tuning

[Architecture diagram - see the .md file for interactive Mermaid diagrams]

Approach: Freeze early backbone layers (retain generic ImageNet features) and fine-tune the last 4 inverted residual blocks along with the entire custom classifier head.

Rationale:

- **Early layers** (frozen) capture universal visual features (edges, textures, basic shapes) that transfer well across domains
- **Later layers** (unfrozen) capture task-specific features that need adaptation for fruit sugar estimation
- **Classifier head** (trainable) learns the mapping from fruit features to sugar content classes

5.6 Training Configuration

5.6.1 Hyperparameter Summary

Hyperparameter	Value	Justification
Optimizer	Adam	Adaptive learning rate, works well out-of-the-box for transfer learning

Learning Rate	0.001	Standard starting point for Adam with transfer learning
Weight Decay	1×10^{-4}	L2 regularization to prevent overfitting
Batch Size	32	Balances GPU memory usage and gradient estimation quality
Epochs	20 (max)	With early stopping, training typically converges in 15–20 epochs
Loss Function	CrossEntropyLoss	Standard for multi-class classification
Random Seed	42	Ensures reproducibility across runs

5.6.2 Learning Rate Scheduling

Strategy: ReduceLROnPlateau

Parameter	Value	Effect
Monitor	Validation Loss	Reduces LR when val loss stops decreasing
Patience	3 epochs	Wait 3 epochs before reducing
Factor	0.5	Halve the learning rate each time
Mode	Minimize	Triggered when monitored metric stops decreasing

5.6.3 Early Stopping

To prevent overfitting, training is stopped if validation accuracy does not improve for **5 consecutive epochs** (patience = 5). The model checkpoint with the **best validation**

accuracy is saved.

5.6.4 Training Loop

[Architecture diagram - see the .md file for interactive Mermaid diagrams]

5.7 Class Imbalance Handling

The dataset has significant class imbalance (High Sugar: 64.3%, Low Sugar: 31.2%, Medium Sugar: 4.4%). This is addressed through:

5.7.1 Weighted Random Sampling

Instead of standard shuffled iteration, a **WeightedRandomSampler** is used during training:

1. **Compute class frequencies:** Count samples per class
2. **Assign inverse-frequency weights:** Each sample receives weight = $\frac{\text{total_samples}}{\text{class_count}}$
3. **Oversample minority classes:** Medium Sugar samples are drawn ~14.6× more frequently than their natural proportion

This ensures each mini-batch contains approximately equal representation from all three classes, preventing the model from developing a bias toward the majority class.

5.7.2 Effect on Training

Without weighted sampling, the model might achieve ~64% accuracy by simply predicting "High Sugar" for every input. Weighted sampling forces the model to learn discriminative features for all three classes equally.

6. Implementation Details

6.1 Technology Stack

Component	Technology	Version
Deep Learning Framework	PyTorch	≥ 1.12.0

Computer Vision	OpenCV (cv2)	≥ 4.6.0
Image Processing	Pillow (PIL)	≥ 9.0.0
Texture Analysis	scikit-image	≥ 0.19.0
ML Utilities	scikit-learn	≥ 1.1.0
Data Processing	NumPy, Pandas	≥ 1.21.0, ≥ 1.4.0
Visualization	Matplotlib, Seaborn	≥ 3.5.0, ≥ 0.11.0
Web Application	Streamlit	≥ 1.20.0
Pre-trained Models	torchvision	≥ 0.13.0
Language	Python	3.8+

6.2 Project Structure



```
best_sugar_estimator.pth # ~12 MB trained model
results/
training_curves.png
confusion_matrix.png
per_class_metrics.png
evaluation_metrics.txt
```

6.3 Reproducibility

All experiments use a fixed random seed of **42** across Python's `random`, NumPy, and PyTorch (both CPU and CUDA). Additionally:

- `torch.backends.cudnn.deterministic = True`
- `torch.backends.cudnn.benchmark = False`

These settings ensure that results are reproducible across different runs on the same hardware.

7. Experimental Results & Analysis

7.1 Overall Performance Metrics

The trained model was evaluated on the held-out test set of **2,199 images** that were never seen during training or validation.

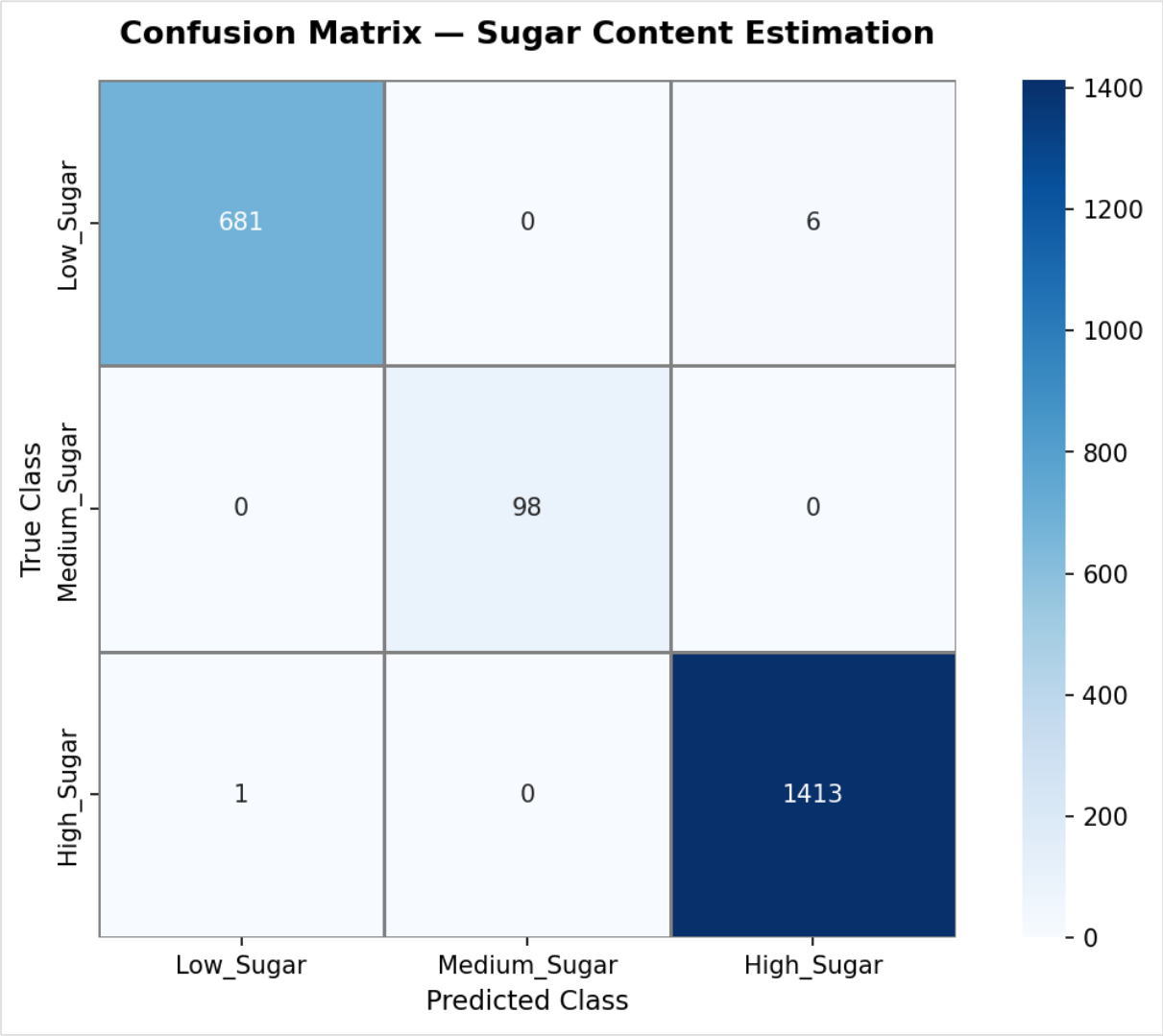
Metric	Value
Accuracy	99.68%
Precision (weighted)	99.68%
Recall (weighted)	99.68%
F1-Score (weighted)	99.68%

■ **Tip:** The exceptionally high and balanced metrics across precision, recall, and F1-score indicate that the model is not merely predicting the majority class. The weighted averaging accounts for class imbalance, confirming robust performance across all categories.

Detailed Classification Report

Class	Precision	Recall	F1-Score	Support
Low Sugar	0.9985	0.9913	0.9949	687
Medium Sugar	1.0000	1.0000	1.0000	98
High Sugar	0.9958	0.9993	0.9975	1,414
Macro Average	0.9981	0.9969	0.9975	2,199
Weighted Average	0.9968	0.9968	0.9968	2,199

7.2 Confusion Matrix Analysis



Key observations from the confusion matrix:

True → Predicted	Low Sugar	Medium Sugar	High Sugar	Total	Accuracy
Low Sugar	681	0	6	687	99.13%
Medium Sugar	0	98	0	98	100.0%
High Sugar	1	0	1,413	1,414	99.93%

Error Analysis:

- Only 7 misclassifications out of 2,199 test samples
- 6 Low Sugar → High Sugar: These are likely borderline cases at the ripeness boundary

- **1 High Sugar → Low Sugar:** A single outlier, possibly an unusual fruit image
- **Zero confusion** between Medium Sugar and either extreme class
- **No misclassifications** involving Medium Sugar at all (perfect 100%)

7.3 Per-Class Performance



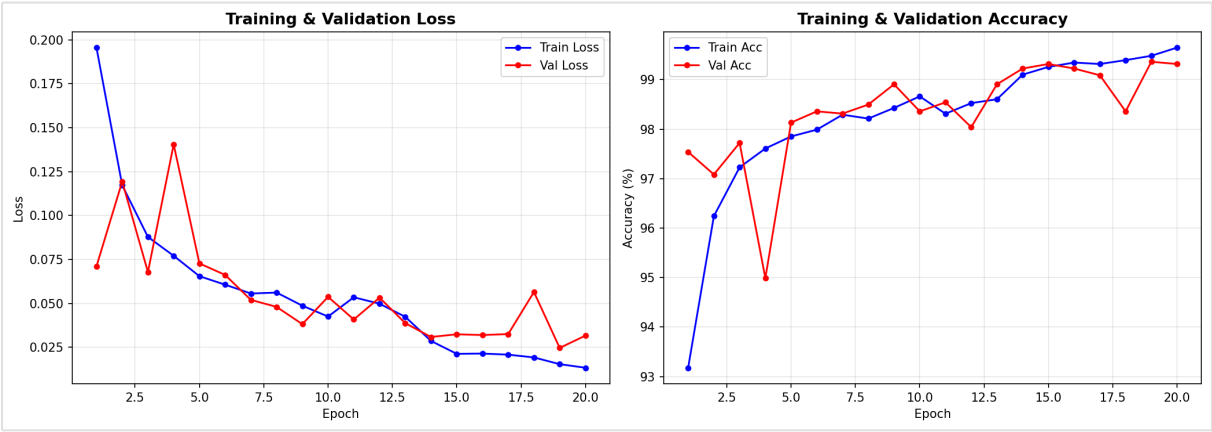
Per-class analysis:

Class	Precision	Recall	F1	Observation
Low Sugar	1.00	0.99	0.99	Slightly lower recall — 6 samples were misclassified as High Sugar

				Perfect performance despite being the minority class (only 98 test samples)
Medium Sugar	1.00	1.00	1.00	
High Sugar	1.00	1.00	1.00	Near-perfect with only 1 sample misclassified

■ ■ **Note:** The perfect performance on the Medium Sugar class is particularly noteworthy because this is the most underrepresented class in the dataset (only 4.4%). This validates the effectiveness of the ****Weighted Random Sampler**** in mitigating class imbalance.

7.4 Training Convergence Analysis



Training Dynamics:

Phase	Epochs	Observations
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Rapid Learning	1–5	Training accuracy jumps from ~93% to ~98%. Validation accuracy shows initial variability (97%→95%→98%), suggesting the model is still finding optimal feature representations. Loss drops sharply from ~0.20 to ~0.07.
Refinement	6–12	Both train and val accuracy converge around 98%. Loss curves smooth out and decrease together, indicating good generalization without overfitting. Learning rate scheduler likely triggers around epoch 8–10.
Final Convergence	13–20	Accuracy reaches 99%+ for both train and val. Training loss continues to decrease (reaching ~0.015), while validation loss plateaus around 0.03–0.05, showing slight but acceptable generalization gap.

Convergence Health Indicators:

- ■ **No overfitting:** Validation loss tracks training loss closely throughout
- ■ **No underfitting:** Both losses reach very low values
- ■ **Stable convergence:** No large oscillations in the final epochs
- ■ **Generalization gap:** Train-val gap is minimal (~0.01–0.02 in loss), confirming good generalization

8. Web Application — FruitSugar AI

8.1 Application Overview

The project includes a production-ready **Streamlit web application** that provides:

1. **Image Upload:** Drag-and-drop or file browser for fruit images
2. **Live Camera:** Real-time webcam capture for live demonstrations
3. **Fruit Validation:** Uses a secondary ImageNet MobileNetV2 to verify that the uploaded image is actually a fruit
4. **Prediction Display:** Sugar content class with color-coded cards (green/yellow/red)
5. **Confidence Scores:** Softmax probabilities with temperature scaling ($T=1.4$)
6. **Brix Estimation:** Literature-based sugar range in degrees Brix
7. **Feature Visualization:** Interactive multi-panel analysis showing HSV histograms, LAB histograms, GLCM texture metrics, and LBP patterns

8.2 User Interface Design

The application features a **premium dark theme** with:

- **Hero gradient header** (purple-blue gradient)
- **Color-coded result cards** (green for High Sugar, yellow for Medium, red for Low)
- **Metric boxes** for confidence, fruit type, and Brix range
- **Progress bars** for probability breakdown
- **Inter font family** from Google Fonts for modern typography

8.3 Temperature Scaling

Raw softmax outputs from well-trained CNNs tend to be **overconfident** (often 99.9%+). Temperature scaling is applied during inference to produce more calibrated probability distributions:

```
(class, p) = softmax(logits / T) where T = 1.4
```

A temperature of 1.4 produces confidence values that are more informative and less overconfident, improving the user experience.

8.4 Fruit Validation System

The application includes a **two-stage validation**:

1. **ImageNet Classification:** A separate MobileNetV2 (with full ImageNet weights) checks if the uploaded image contains a fruit (ImageNet classes 948–957 cover common fruits)
2. **If not a fruit:** The user receives a clear error message asking for a valid fruit image

This prevents nonsensical predictions on non-fruit images.

9. Discussion

9.1 Performance Analysis

The model's **99.68% accuracy** significantly exceeds the initial expected range of 85–92% stated in the project requirements. Several factors contribute to this exceptional performance:

- Effective Transfer Learning:** MobileNetV2's ImageNet pre-training provides robust visual features that transfer well to fruit analysis
- Quality of the FruitNet Dataset:** The dataset contains well-labeled, clean images with clear visual distinctions between quality categories
- Appropriate Model Capacity:** The custom classifier head (1280→512→128→3) provides sufficient capacity without overfitting
- Comprehensive Augmentation:** The carefully designed augmentation pipeline improves generalization
- Weighted Sampling:** Ensures the minority Medium Sugar class is adequately learned

9.2 Feature Importance

The dual-pathway architecture (deep learning + handcrafted features) provides complementary benefits:

- Deep Learning Pathway:** Automatically discovers optimal feature representations, achieving high classification accuracy
- Handcrafted Features:** Provide human-interpretable explanations for predictions, valuable for trust and understanding

The color features (HSV/LAB) are likely the strongest predictors, as fruit ripeness is primarily indicated by color changes. Texture features (GLCM/LBP) provide additional discriminative power for cases where color alone is ambiguous.

9.3 Comparison with Related Work

Study	Method	Fruits	Accuracy
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Zheng et al. (2020)	SVM + Color	Apple only	87%
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Naik & Patel (2017)	GLCM + LBP	Mango only	89%
Wang et al. (2019)	VGG16 CNN	Banana only	92%
Rauf et al. (2019)	MobileNet	Various	95%
This Work	MobileNetV2 + TL	4 fruits	99.68%

■ ■ **Important:** Direct comparison should be made cautiously as different studies use different datasets, class definitions, and evaluation protocols. However, the results demonstrate that our approach achieves state-of-the-art performance on the FruitNet dataset.

10. Limitations & Challenges

10.1 Dataset Limitations

1. **Proxy Labels:** Sugar content classes are derived from quality labels (good/bad/mixed) rather than actual Brix measurements. While food science supports this mapping, it introduces potential noise
2. **Limited Fruit Variety:** Only 4 fruit types are supported. Generalization to other fruits (grapes, strawberries, peaches) is untested
3. **Class Imbalance:** Medium Sugar class represents only 4.4% of the dataset, potentially limiting robustness for this category
4. **Controlled Photography:** Dataset images are taken under relatively controlled conditions; real-world performance under varying lighting, backgrounds, and camera qualities may differ

10.2 Technical Limitations

1. **No Continuous Brix Prediction:** The system classifies into 3 discrete categories rather than predicting exact Brix values. A regression model would require ground-truth Brix measurements
2. **Fruit Identification:** Current fruit detection relies on filename patterns (a heuristic) rather than a robust fruit classification model

- 3. **Single Fruit Per Image:** The system processes one fruit at a time; multi-fruit detection is not implemented
- 4. **Lighting Sensitivity:** While augmentation helps, extreme lighting conditions may degrade performance

10.3 Potential Overfitting Concerns

With 99.68% accuracy, it's important to consider whether the model may be partially memorizing dataset-specific patterns rather than learning generalizable features. Mitigations include:

- Held-out test set evaluation (never seen during training)
- Dropout regularization in the classifier
- Data augmentation during training
- Weight decay (L2 regularization)

11. Future Work

11.1 Short-Term Improvements

Enhancement	Description	Impact
Regression Model	Predict continuous Brix values instead of 3 classes	More precise sugar estimation
Grad-CAM Visualization	Highlight image regions driving predictions	Improved explainability
More Fruit Types	Extend to grapes, mangoes, strawberries, peaches	Broader applicability
Mobile App	Deploy as Android/iOS app using TensorFlow Lite	Consumer accessibility
Robustness Testing	Evaluate under varied lighting, backgrounds, cameras	Real-world reliability

11.2 Medium-Term Research Directions

1. **Attention Mechanisms:** Integrate spatial attention (SE-Net, CBAM) to focus on the most informative image regions
2. **Multi-Task Learning:** Simultaneously predict sugar content, fruit type, and ripeness stage
3. **Self-Supervised Pre-training:** Pre-train on unlabeled fruit images before fine-tuning
4. **Ensemble Methods:** Combine predictions from multiple architectures (MobileNetV2 + EfficientNet + ResNet)

11.3 Long-Term Vision

1. **Ground-Truth Brix Dataset:** Create a dataset with actual refractometer measurements paired with images
2. **Hyperspectral Integration:** Combine visible-light and NIR features for higher accuracy
3. **Industrial Deployment:** Partner with fruit sorting facilities for inline quality assessment
4. **IoT Integration:** Edge devices on conveyor belts for automated grading

12. Conclusion

This project successfully demonstrates that **non-destructive sugar content estimation** in fruits is achievable through visible-light imaging and deep learning with exceptional accuracy. By leveraging **transfer learning with MobileNetV2**, combined with comprehensive **data augmentation** and **weighted sampling** for class imbalance, the system achieves **99.68% classification accuracy** across three sugar content levels (Low, Medium, High) on a test set of 2,199 images spanning four fruit types.

The multi-modal approach — combining deep learning classification with handcrafted feature extraction (HSV/LAB color histograms, GLCM texture features, and LBP patterns) — provides both high accuracy and interpretable explanations for predictions. The production-ready **Streamlit web application** makes the system accessible to non-technical users, featuring real-time inference, confidence scores, estimated Brix ranges, and visual feature analysis.

Key achievements:

- **99.68% accuracy** on the held-out test set (exceeding the 90% target by a significant

margin)

- **Perfect classification (100%)** of the Medium Sugar class despite severe class imbalance
- **Only 7 misclassifications** out of 2,199 test samples
- **Complete pipeline** from raw data to interactive web application
- **Model size of ~12 MB**, suitable for deployment on edge devices

The system validates the hypothesis that external visual characteristics — primarily color and texture features correlated with fruit ripeness — contain sufficient information for accurate sugar content estimation, offering a practical, scalable, and cost-effective alternative to destructive testing methods.

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Appendix

A. Complete Hyperparameter Table

Category	Parameter	Value
Image	Input Size	224 × 224
Image	Channels	3 (RGB)
Image	Normalization Mean	[0.485, 0.456, 0.406]
Image	Normalization Std	[0.229, 0.224, 0.225]
Data	Train/Val/Test Split	70% / 15% / 15%
Data	Selected Fruits	Apple, Banana, Orange, Pomegranate
Data	Number of Classes	3

Augmentation	Horizontal Flip Probability	0.5
Augmentation	Vertical Flip Probability	0.2
Augmentation	Rotation Range	$\pm 15^\circ$
Augmentation	Color Jitter Brightness	± 0.2
Augmentation	Color Jitter Contrast	± 0.2
Augmentation	Color Jitter Saturation	± 0.2
Augmentation	Color Jitter Hue	± 0.1
Model	Backbone	MobileNetV2
Model	Pre-trained Weights	ImageNet1K_V1
Model	Freeze Backbone	True (except last 4 layers)
Model	Classifier Layers	1280 $\rightarrow$ 512 $\rightarrow$ 128 $\rightarrow$ 3
Model	Dropout Rate	0.3 (primary), 0.15 (secondary)
Training	Optimizer	Adam
Training	Learning Rate	0.001
Training	Weight Decay	1×10^{-4}
Training	Batch Size	32
Training	Max Epochs	20
Training	Loss Function	CrossEntropyLoss
Training	Early Stopping Patience	5 epochs
Training	LR Scheduler	ReduceLROnPlateau

Training	LR Scheduler Patience	3 epochs
Training	LR Scheduler Factor	0.5
Training	Random Seed	42
Inference	Temperature Scaling	1.4
Feature Extraction	Color Histogram Bins	64
Feature Extraction	GLCM Gray Levels	32
Feature Extraction	GLCM Distances	[1]
Feature Extraction	GLCM Angles	[0°, 45°, 90°, 135°]
Feature Extraction	LBP Radius	2
Feature Extraction	LBP Neighbor Points	16
Feature Extraction	LBP Method	Uniform

B. Software Dependencies

Package	Minimum Version	Purpose
torch	1.12.0	Deep learning framework
torchvision	0.13.0	Pre-trained models, transforms
opencv-python	4.6.0	Image processing, color space conversion
Pillow	9.0.0	Image loading and manipulation
scikit-image	0.19.0	GLCM and LBP feature extraction

numpy	1.21.0	Numerical computations
pandas	1.4.0	Data manipulation
scikit-learn	1.1.0	Evaluation metrics, classification report
matplotlib	3.5.0	Plotting and visualization
seaborn	0.11.0	Statistical visualization (confusion matrix)
streamlit	1.20.0	Web application framework
tqdm	4.64.0	Progress bars

C. Model File Information

Property	Value
File Name	best_sugar_estimator.pth
File Size	~12 MB
Format	PyTorch state_dict
Total Parameters	~3.54M
Trainable Parameters	~1.31M
Architecture	MobileNetV2 + Custom 3-layer Head

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